

Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

- A. It activates TLR5 on the surface of macrophages.
- B. It primarily induces the Th1-cell response.
- C. Peptidoglycan is its major antigen that induces an immune response.
- D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

< reasoning process here > The patient presents with symptoms consistent with community-acquired pneumonia (CAP), including sore throat, fever, cough, and physical exam findings of oropharyngeal erythema and moist rales. The chest X-ray shows patchy reticular opacities in the perihilar regions, which are characteristic of bacterial pneumonia. The most common causative agent of community-acquired pneumonia in adults is Streptococcus pneumoniae, a Gram-positive bacterium.

Streptococcus pneumoniae is known to induce a strong immune response, particularly through the activation of innate immune receptors such as Toll-like receptors (TLRs). However, TLR5 recognizes flagellin, which is found in Gram-negative bacteria, not Gram-positive bacteria like S. pneumoniae. Therefore, option A is incorrect.

S. pneumoniae primarily triggers a Th1-type immune response, which is important for fighting intracellular pathogens and bacterial infections. Th1 responses involve the production of interferons and cytokines like IL-12 and IFN-γ, which are critical in activating macrophages and cytotoxic T cells. This aligns with the immune response to bacterial pneumonia.

Option C, stating that peptidoglycan is the major antigen, is partially correct in that peptidoglycan is a component of bacterial cell walls and can be recognized by immune cells. However, peptidoglycan is not the "major antigen" that defines the immune response in the context of the question. It's a structural component, and while it is recognized by NOD-like receptors in macrophages, it's not the best description of the immune response elicited.

Option D, encapsulation as a mechanism to evade immunity, is true for S. pneumoniae (the capsule prevents phagocytosis), but the question asks for the immune response elicited, not immune evasion mechanisms.

Therefore, the best answer describing the immune response is the Th1 response, which is critical in clearing bacterial infections like pneumonia.

< answer > B. It primarily induces the Th1-cell response. </answer >